CHAPTER 4

WEB TECHNOLOGY

CHAPTER ESSENCE AT A GLANCE:
Analyze client-server web architectures, DNS domain name resolution, HTTP request-response cycles, static vs dynamic pages, and multi-tier web stacks.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Client-Server Communication:** Distributed web computing pairs client browsers that request services over standard transmission protocols with remote web servers that process incoming requests and return structured responses.

• **Static and Dynamic Web Architecture:** Static web pages deliver identical pre-built layout documents directly from server storage, whereas dynamic pages generate tailored, database-driven content in real time upon each user request.

• **Dual Scripting Responsibilities:** Client-side scripts execute within the user browser to validate inputs and animate interfaces, whereas server-side scripts run privately on the host server to manage databases and execute business logic securely.

• **Standard Protocols and Dedicated Ports:** Standard communication rules govern internet traffic, designating port eighty for regular web traffic, port four hundred forty-three for encrypted secure web connections, and port twenty-one for file transfers.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

$$\text{HTTP} = 80 \quad | \quad \text{HTTPS} = 443 \quad | \quad \text{DNS} = 53 \quad | \quad \text{FTP} = 21$$

Standard internet protocol dedicated port assignments

$$\text{URL} = \text{Protocol} + \text{Domain Name} + \text{Port} + \text{Resource Path}$$

Uniform Resource Locator standard anatomical structure

$$\text{Throughput} = \frac{\text{Delivered Payload Bytes}}{\text{Elapsed Transfer Time}}$$

Network transmission bandwidth performance relation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Client-side scripts run locally on the client browser for validation, whereas server-side scripts execute on the remote host for data access.

• **Alert:** HTTPS encrypts payload data using TLS and SSL over port 443, unlike unencrypted standard HTTP traffic over port 80.

• **Alert:** DNS converts human-readable domain names into 32-bit IPv4 or 128-bit IPv6 numerical addresses.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain client-server architecture in web technology and the communication workflow. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Architecture):** Client-server architecture is a distributed computing structure where the client (web browser) initiates requests for services/resources over HTTP/HTTPS, and the web server processes requests, executes scripts, accesses databases, and serves responses back to client. **[3 Marks]**

• **Key Exam Result:** Client requests resources over HTTP; Web server processes and serves responses

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

2. Explain the roles and examples of Web Browsers and Web Servers. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Web Browser):** Client-side application software used to locate, retrieve, render, and display web pages to users. Examples: Google Chrome, Mozilla Firefox, Apple Safari. **[1 Mark]**

• **Step 2 (Web Server):** Server software/hardware running continuously to listen for incoming HTTP requests, process backend logic, and deliver web pages to clients. Examples: Apache HTTP Server, Nginx, Microsoft IIS. **[1 Mark]**

• **Key Exam Result:** Browser: Chrome, Firefox (client) | Server: Apache, Nginx (host)

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

3. Differentiate between Static Web Pages and Dynamic Web Pages (four points). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Static Web Pages):** Content is pre-built and fixed; identical HTML delivered to all users; fast loading; requires no backend database; modified only by directly editing HTML source files. **[1.5 Marks]**

• **Step 2 (Dynamic Web Pages):** Content generated on-the-fly in response to user input, session state, or database queries; customized for each user; interactive; uses server-side scripts (PHP, Python) and databases (MySQL). **[1.5 Marks]**

• **Key Exam Result:** Static (fixed content, no database) vs Dynamic (real-time generated, interactive)

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

4. Differentiate between Client-Side Scripting and Server-Side Scripting. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Client-Side Scripting):** Executed directly within user's web browser; used for form validation, UI animations, and immediate event handling without round-trips to server. Example: JavaScript. **[1.5 Marks]**

• **Step 2 (Server-Side Scripting):** Executed on web server before HTML is sent to browser; handles database queries, business logic, authentication, and security. Source code is hidden from user. Examples: PHP, Python (Django), Node.js. **[1.5 Marks]**

• **Key Exam Result:** Client-side: JavaScript (validation/UI) | Server-side: PHP (database/logic)

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

5. Explain HTTP and HTTPS protocols. Why is HTTPS preferred for secure web communication? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Protocols):** HTTP (Hypertext Transfer Protocol): Application protocol for data transfer on web; unencrypted plain text (port 80). HTTPS (HTTP Secure): Encrypted communication using SSL-TLS (port 443), protecting against eavesdropping and data tampering. FTP (File Transfer Protocol): Used for transferring bulk files between client and server (port 21). TCP-IP: Underlying suite ensuring reliable, error-checked packet transmission. **[3 Marks]**

• **Key Exam Result:** HTTP (unencrypted), HTTPS (SSL-TLS encrypted), FTP (file upload/download)

$$\text{HTTP} = 80 \quad | \quad \text{HTTPS} = 443 \quad | \quad \text{DNS} = 53 \quad | \quad \text{FTP} = 21$$

Standard internet protocol dedicated port assignments

6. Explain the architecture and working of search engines: Web Crawling, Indexing, and Searching. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Search Engine Architecture):** 1. Web Crawlers or Spiders: Automated bots (e.g. Googlebot) that continuously traverse web hyperlinks and download web pages. 2. Indexing Software: Parses downloaded text, extracts keywords, and stores data in massive inverted index databases. 3. Query Processor & Ranking Engine: Evaluates user search terms, matches indexed data, and ranks results by relevance and PageRank algorithms. **[3 Marks]**

• **Key Exam Result:** Crawling (downloading), Indexing (categorizing), Ranking (serving results)

$$\text{Client (Browser)} \underset{\text{Response (HTML)}}{\overset{\text{Request (HTTP)}}{\rightleftharpoons}} \text{Web Server}$$

Client-server request and response communication cycle

7. Explain the systematic sequence of events when a user enters a URL into a browser until the page loads. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Systematic Sequence):** 1. User enters URL in browser. 2. Browser requests IP address from DNS. 3. Three-way TCP handshake established with server (SYN, SYN-ACK, ACK). 4. Browser sends HTTP GET request. 5. Server processes request and returns HTTP 200 OK with HTML. 6. Browser parses HTML, fetches CSS/JS/images, and renders final DOM on screen. **[3 Marks]**

• **Key Exam Result:** DNS lookup --> TCP handshake --> HTTP GET --> Server response --> DOM render

$$\text{URL} = \text{Protocol} + \text{Domain Name} + \text{Port} + \text{Resource Path}$$

Uniform Resource Locator standard anatomical structure